Highlights

Laser-Assisted Nuclear Thermal Propulsion for Cislunar Logistics: A Monte Carlo Power-Closure Assessment

- Reaching 1300-1500 s needs laser power of 2.2-3.6 times the reactor power
- A 1-50 MW laser on a 100-500 MWt reactor gives at most 1,000 s
- Median IMLEO is 1.7 times that of conventional NTP; 0% of draws reach -40%
- Beam capture collapses beyond a few 10^4 km, ruling out relay beaming
- Laser nodes outweigh the tug and a single laser fault erodes the benefit